

BLOOD TRIBE AGDM, AB, Canada

WMO: 715170

Lat: **49.5588N**Lon: **113.0633W**Elev: **3278**StdP: **13.04**Time Zone: **-7.00 (NAM)**Period: **04-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-19.1	-12.9	-25.1	1.5	-18.9	-18.3	2.3	-12.7	43.3	35.6	41.1	36.9	7.5	230	0.748

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.5	87.3	61.1	83.8	60.4	80.5	59.7	65.9	78.7	63.7	77.2	61.7	75.8	10.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	91.3	72.2	58.7	83.3	69.9	56.5	76.7	66.7	32.5	78.9	30.8	77.2	29.2	75.9	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.6	32.3	28.5	DB	-28.8	93.5	6.7	4.1	-33.6	96.4	-37.6	98.8	-41.4	101.1	-46.3	104.1	(4)
(5)			WB	-29.0	70.6	6.7	2.9	-33.8	72.7	-37.7	74.4	-41.5	76.0	-46.3	78.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.8	23.5	21.1	31.4	40.9	50.8	57.7	63.9	62.0	54.1	42.5	31.4	20.3		
	DBStd	18.89	15.49	15.67	13.18	8.78	7.75	5.65	5.34	6.02	8.23	9.73	13.22	15.47		
	HDD50	4388	821	811	579	289	85	6	0	2	50	264	562	919		
	HDD65	8589	1285	1230	1043	724	443	229	84	128	333	697	1009	1384		
	CDD50	1379	1	0	1	15	110	236	431	376	175	31	3	0		
	CDD65	105	0	0	0	0	2	9	50	37	7	0	0	0		
	CDH74	2687	0	0	0	13	129	244	1047	923	323	8	0	0		
	CDH80	813	0	0	0	1	20	46	346	314	86	0	0	0		
(14) Wind	WSAvg	11.9	14.8	12.3	12.6	12.6	11.2	11.1	9.0	8.7	10.3	12.3	13.9	13.9		
(15) Precipitation	PrecAvg	17.2	0.7	0.5	0.9	1.4	2.6	3.8	1.7	1.5	1.5	0.8	0.7	0.7		
	PrecMax	27.0	1.2	1.6	2.0	3.2	4.9	12.1	5.9	3.8	5.2	2.9	1.5	1.5		
	PrecMin	8.1	0.3	0.0	0.1	0.4	0.5	0.7	0.0	0.0	0.2	0.1	0.1	0.1		
	PrecStd	4.5	0.2	0.4	0.5	0.7	1.2	2.5	1.3	0.9	1.1	0.6	0.4	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.0	55.8	62.8	75.4	82.9	86.4	92.5	91.8	87.3	73.9	64.0	52.5		
		MCWB	43.0	41.8	45.4	51.8	56.0	63.7	62.6	60.3	58.3	52.6	47.6	40.6		
	2%	DB	49.6	50.2	57.6	67.4	77.6	80.1	87.5	87.3	82.4	68.2	57.0	47.6		
		MCWB	40.2	39.2	43.2	48.2	54.7	60.3	62.5	60.1	57.0	50.6	44.7	38.9		
	5%	DB	45.2	45.8	53.8	62.5	73.0	76.4	83.9	83.3	77.5	62.8	52.1	43.6		
		MCWB	37.2	36.8	41.7	45.6	52.8	58.7	62.5	59.5	55.8	48.4	42.0	36.4		
	10%	DB	41.9	41.8	49.4	57.2	68.2	72.4	80.0	79.4	71.9	57.8	48.2	40.5		
		MCWB	35.3	34.5	39.2	43.5	50.7	56.6	61.9	59.1	53.9	46.0	39.5	34.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.3	43.1	47.3	52.2	57.8	67.2	70.6	66.1	61.1	54.9	49.7	43.0		
		MCDB	53.1	53.4	59.8	72.2	77.1	79.6	82.3	79.2	78.1	70.4	61.9	50.8		
	2%	WB	40.3	39.8	44.5	49.0	55.7	63.2	67.2	63.8	58.4	51.9	45.6	39.4		
		MCDB	48.5	49.3	56.2	65.4	74.9	76.6	79.7	78.5	77.1	65.8	56.1	46.0		
	5%	WB	37.7	37.0	42.1	46.4	53.7	60.1	65.0	61.9	56.7	49.2	42.5	36.7		
		MCDB	44.4	45.0	52.5	61.3	71.0	72.7	77.7	77.4	74.9	61.2	51.3	43.1		
	10%	WB	35.7	34.8	39.5	43.9	51.8	57.9	63.1	60.2	54.5	46.5	39.9	34.4		
		MCDB	41.5	41.4	48.9	56.1	66.0	69.7	76.1	75.7	71.1	57.0	47.7	40.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.7	20.9	21.4	23.5	24.3	23.5	28.5	29.1	26.5	22.9	19.8	18.8		
		MCDBR	19.8	24.0	27.8	33.3	33.2	31.5	35.8	38.0	37.5	30.3	24.8	19.4		
		MCWBR	12.7	15.0	16.0	16.5	14.6	15.1	16.4	15.6	15.9	15.4	14.6	13.4		
	5% WB	MCDBR	19.6	22.9	26.3	31.0	30.6	28.3	31.0	32.4	34.4	28.8	24.0	19.5		
		MCWBR	13.3	15.1	15.9	15.9	13.7	14.7	16.6	15.4	15.3	15.2	14.7	14.1		
(40) Clear-Sky Solar Irradiance	taub		0.248	0.258	0.280	0.319	0.331	0.327	0.332	0.348	0.311	0.280	0.256	0.232		
	taud		2.440	2.467	2.484	2.386	2.411	2.466	2.462	2.395	2.472	2.538	2.500	2.446		
	Ebn at Noon		265	289	299	295	294	295	292	282	283	277	260	257		
	Edh at Noon		19	23	28	34	35	34	33	34	28	22	18	16		
(44) All-Sky Solar Radiation	RadAvg		394	712	1110	1536	1830	2041	2246	1784	1267	790	443	314		
	RadStd		33	54	94	103	156	150	117	109	104	79	35	30		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page